

Developments

SFINCS has continually being developed since 2017, and many great features have been added over the years.

Development status

See here a schematic overview of the SFINCS development status at November 2025, at the time of the v2.3.0 mt. Faber release. Indicated are new functionality for SFINCS itself (core), and model setup/post-processing using HydroMT-SFINCS (Python).

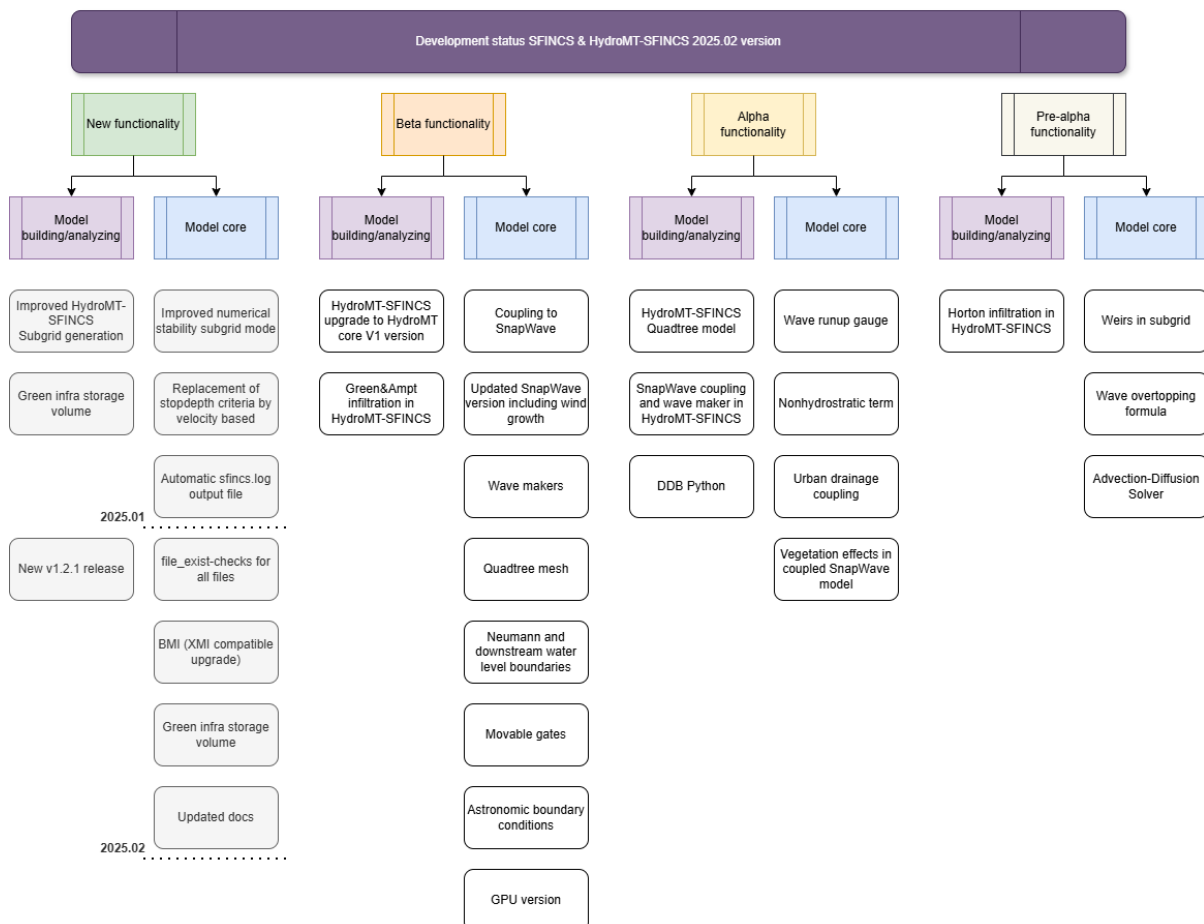


Fig. 10 Overview of SFINCS development status 2025.02 Release

Known issues

Is your app OWASP compliant? Get the latest guidelines for implementing robust mobile app security.

stable ▼

Ads by
EthicalAds

Close Ad

- If you are still using restartfiles from SFINCS 2023 Cauberg release or older, you should remake your restartfiles to overcome potential issues with running the col d'Eze, mt Faber or future releases!
- The BMI implementation in SFINCS is up to date with XMI (BMI + extensions - Hughes et al. 2022), to be used with 'xmipy' (<https://github.com/Deltares/xmipy>) and related functions (<https://deltares.github.io/xmipy/xmipy.html>), which is however not up to date with the latests CSDMS standard BMI implementation 2.0.
- The combination of netspwfile with large difference in reference time between the spiderweb and the SFINCS simulation itself, might not run correctly in the Docker version. Use the ascii spwfile input or the Windows build executable which work correctly.
- Docker GPU version of Deltares latest is not fully functional, and therefore removed from the repo. If you'd want to use the GPU version of SFINCS, get in touch to set up a collaboration.
- Issue in SFINCS v2.3.0 mt Faber Release regarding Curve Number infiltration if storecumprcp = 0 (default), then infiltration is not processed correctly and can result to unrealistic results! Simple solution for now: put storecumprcp = 1 when using this infiltration option. This issue is already fixed in SFINCS main, and therefore also in the next official release!

Releases Changelog

Official open source version 2025.02: v2.3.0 mt. Faber release

The second official 2025 release of SFINCS, the v2.3.0 mt Faber release, 'Modernizing Tasks in Flood Assessments for Better Emergency Response', is now available!

This contains open access to the source code from Github: https://github.com/Deltares/SFINCS/releases/tag/v2.3.0_mt_Faber_release.

As pre-compiled Windows executable:

<https://download.deltares.nl/en/sfincs/>

As Docker container:

`docker pull deltares/sfincs-cpu:sfincs-v2.3.0-mt-Faber-Release`

Changes:

The code consists of all functionality of the 2025.01 'v2.2.0 col d'Eze' release, v

are not found. SFINCS will stop if a declared filename in sfincs.inp is not found when read in.

- Storage volume option for green infrastructure “volfile”, including HydroMT-SFINCS setup function.
- Flag to turn on writing away the time stamp that the maximum water elevation during simulation occurred (storetzsmax = 1).
- Flag to turn on writing away the volume in the subgrid cells (storezvolume = 1).
- Flag to turn on writing away the volume in the storage volume cells (storestoragevolume = 1).
- For thin dams now also the snapped x&y locations are written to the his-file, as for weirfiles.
- Added infiltration to logfile processes section.
- Added explanation on how to use the check valve option in drainage structures in the documentation.
- Add description for utmzone input in case of spiderwebs in latlon.
- New Python setup tools HydroMT-SFINCS release > recommended to use this new version (v1.2.1) instead of the last release!

Bugfixes:

- Fixed bug with restartfile where fluxes were not read in fully correctly due to mismatch length of variable (dummy +1). NOTE - if you are still using restartfiles from SFINCS 2023 Cauberg release or older, you should remake your restartfiles to overcome potential issues!
- Fixed bug in quadtree variable dyrinvc use and definition.
- Fixed bug when spw_merge_frac is different from default 0.5.
- Fixed bug in use of SnapWave with IG balance in relation to definition of inner cells

Advanced user options - currently as alpha/beta functionality:

- NOTE - please contact Deltares-SFINCS group in case you want to use any of this functionality.
- Update of the integrated SnapWave solver to be consistent the latest standalone version (<https://doi.org/10.5194/egusphere-2025-492>).
- Option to include or exclude wind from SFINCS to SnapWave coupling, to include wind growth on wave heights or not.
- Added Neumann water level boundary condition option.
- Added downstream riverine water level boundary condition option.
- Added runoff gauge option.
- Added variations on drainage structures with moveable gates, triggered by either water level exceedance or provided time.
- Added option to use bcafile astronomical boundary conditions to internally generate tidal water levels at msk=2 boundary similar to Delft3D.

- Upgraded GPU implementation for consistent results with CPU version.
- Upgraded BMI implementation exposing more variables to be altered.

Official open source version 2025.01: v2.2.0 col d'Eze release

The first official 2025 release of SFINCS, the v2.2.0 col d'Eze release, 'Easing over bumps in fast compound flood modelling together', is now available!

This contains open access to the source code from Github: https://github.com/Deltares/SFINCS/releases/tag/v2.2.0_col_dEze_release.

As pre-compiled Windows executable:

<https://download.deltares.nl/en/sfincs/>

As Docker container:

```
docker pull deltares/sfincs-cpu:sfincs-v2.2.0-col-dEze-Release
```

Changes:

The code consists of all functionality of the v2.1.1 release, with the following major changes/additions:

- Improvements on numerical stability. Achieved through additions of `uvmax`, `hmin_cfl`, `uvlim`, `slopelim`, `advlim` (see below) to replace the old stability criteria using 'stopdepth' with square root of `gh` (removed, see below). Impacts both regular gridded as subgrid model simulations.
- Improvements subgrid mode through added checks on input file, subgrid mode consistent with SFINCS subgrid paper: van Ormondt et al. (2025): <https://doi.org/10.5194/gmd-18-843-2025>.
- Addition of automatic creation of a 'sfincs.log' output file. Implemented so that only a shortened output is send to screen (or in 'sfincs_log.txt' as in our examples using 'run.bat' batchfile), and a longer more elaborate version to the 'sfincs.log' file. NOTE - in the run.bat batchfile you can therefore not specify '/sfincs.exe>sfincs.log' anymore, since that conflicts with the automatically generated file. Any other name (as `sfincs_log.txt` in our examples) will do, or don't specify an output file for the messages to the screen at all!
- Added more clear overview of what parameters are turned on or off by displaying a summarizing table in the log file and to screen messages (e.g. `coriolis`, also see below).
- Added option to set how frequent to show progress of SFINCS in terms of % and time remaining in `sfincs inn` using `percentage_done` (default = 5%). So can also be

 [stable](#) ▼

version (<https://doi.org/10.5194/egusphere-2025-492>).

- New Python setup tools HydroMT-SFINCS release > highly recommended to use this new version instead of the last release (v1.1.0)!
- Update of executable license from 'Deltares free trial copy' to 'Deltares freeware license', to be accepted upon downloading from portal (same as before). For details always read the full license (LICENSING CONDITIONS DELTARES FREWARE EXECUTABLE.txt).

Detailed overview additions/changes:

- stopdepth - REMOVED in SFINCS v2.2.0, replaced by 'uvmax' to determine possible instabilities based on flow velocities rather than maximum water depth!
- uvmax - possibility to set maximum flux velocity (default 1000 m/s), used to determine minimum timestep, below which simulation is classified as unstable and stopped. Replaces 'stopdepth'.
- hmin_cfl - possibility to set minimum water depth to determine maximum timestep using CFL-conditions.
- uvlm - possibility to limit flux velocity (default 10 m/s).
- slopelim - possibility to apply slope limiter to dzdx (turned off by default, by setting to 9999.9).
- advlim - updated use of the advection limiter, new default is 1.0, whereby limiter is turned on by default.
- coriolis - clarification of use in model and logfile: for projected coordinate systems only turned on if a latitude is provided other than 0 (default, latitude = 0.0, means no coriolis terms used in momentum equation). For large scale applications on spherical grid, the coriolis term is turned on by default.
- waterlevel 'zs' and volume 'z_volume' internal variables in SFINCS kernel are now stored as double precision.

Bugfixes:

- Bugfix structures (pump, culvert, valve) so they cannot fall dry to NaNs in case the sink term cell becomes fully dry.
- Bugfix advection scheme. Influence on real world application cases in our testbed is small/negligible.

Advanced user options - currently as alpha/beta functionality:

- NOTE - please contact Deltares-SFINCS group in case you want to use any of this functionality.
- Ungraded BMI implementation to be compliant with that of XMI (BMI + ext

- Nonhydrostatic pressure correction (for tsunami wave modelling), keywords nonh = yes (default no) and 'nh_tstop', 'nh_fnudge', 'nh_tol', 'nh_itermax'. Also added option to specify 'nonh_mask' to turn on nonh correction only in part of the domain.
- Added lookup table for $h^{(7/3)}$ term in momentum equation. Potentially faster. Keyword h73table = 1/yes/true (default: false).
- Added option to store subgrid hmean (rather than zs - z_zmin), keyword storehmean = yes (default no)

Official open source version 2024.01: v2.1.1 Dollerup release

The first official 2024 release of SFINCS, the v2.1.1 Dollerup release, 'Conquering challenges in compound flood modelling together', is now available!

This contains open access to the source code from Github: https://github.com/Deltares/SFINCS/releases/tag/v2.1.1_Dollerup_release.

As pre-compiled Windows executable:

<https://download.deltares.nl/en/sfincs/>

As Docker container:

`docker pull deltares/sfincs-cpu:sfincs-v2.1.1-Dollerup-Release`

Changes:

The code consists of all functionality of the v2.0.3 release, with the following major changes/additions:

Added functionality:

- New advection scheme 'advection_scheme = upw1' > new default! Old implementation still available as 'advection_scheme = original'.
- Option to include 2D component of friction term 'friction2d = true' > new default! Old implementation still available as 'friction2d = false'.
- New Python setup tools HydroMT-SFINCS release v1.1.0 > highly recommended to use this new version!
- New subgrid methodology including wet fraction as in Van Ormondt et al. (2024, in review), only available through net netcdf input file through new HydroMT-SFINCS version recommended! Old implementation still available when providing the origin

wave energy balance following Leijnse et al. (2024).

- New recommended default combination that with new advection scheme: $\alpha=0.50$, $\theta=1.0$, $\text{advection}=1$ (is now always 2D), $\text{viscosity}=1$.

Other changes:

- Improved stability for cell that from being dry becomes wet.
- Extra output variable to netcdf output files 'status', for easy checking whether a batch of simulations have all run successfully: a value of 0 means no error
- Added option to specify in case of a 1 layer model (either quadtree input, or original input), whether you want the output as mesh quadtree netcdf (1), or regular gridded netcdf (0, default). Keyword: `regular_output_on_mesh` (0, default, =logical)
- `netspwfile` input for precipitation should be absolute atmospheric pressure, not the pressure drop.
- Upgrade Windows compiler to IFX 2024.
- Upgraded Netcdf version to newer native versions of Unidata: netCDF 4.9.2 & netcdf-fortran-4.6.1
- Updated documentation.
- Added tests in skillbed report.

Official open source version Q4 2023: v2.0.3 Cauberg release

The last official 2023 release of SFINCS for Q4, the v2.0.3 Cauberg release, 'Climbing hills of challenges in compound flood modelling', is now available!

This contains open access to the source code from Github: https://github.com/Deltares/SFINCS/releases/tag/v2.0.3_Cauberg_release.

As pre-compiled Windows executable:

<https://download.deltares.nl/en/sfincs/>

As Docker container:

```
docker pull deltares/sfincs-cpu:sfincs-v2.0.3-Cauberg
```

Changes:

The code consists of all functionality of the v2.0.2 release, with the following changes/additions:

- Netcdf4 compliant output, for reduced file sizes
- Green & Ampt infiltration method
- Horton infiltration method
- Restartfiles including infiltration state, for the case of Curve Number with Recovery, Green&Ampt and Horton
- Netcdf Spiderweb input option
- Netcdf src+dis input option

Other changes:

- Improved culvert and drainage pipe formulations
- Updated documentation
- Added tests in skillbed report
- Compliance with new Python setup tools HydroMT-SFINCS release v1.0.2

Official open source version Q2 2023: v2.0.2 Blockhaus release

As the first out of 2 official 2023 releases, the v2.0.2 Blockhaus release, 'Smoothly cycling over challenges in compound flood modelling', is now available as Windows executable: <https://download.deltares.nl/en/sfincs>

And Docker container: `docker pull deltares/sfincs-cpu:sfincs-v2.0.2-Blockhaus`

This contains open access to the source code from Github: <https://github.com/Deltares/SFINCS/releases/tag/v2.0.2>.

The code consists of all functionality of the v2.0.0 release, with the following changes/additions:

- Potentially breaking change: flipped x&y coordinates in Netcdf map output to be Sgrid compliant. Note; might impact Matlab/Python post-processing scripts (fixed in new HydroMT-SFINCS release v1.1.0)
- Improved 2D component of advection scheme
- Option to not use rainfall in spiderweb, keyword: `usespwprecip = 0`
- The x&y-coordinates of input weirfiles as snapped on grid internally in SFINCS are now written to the `sfincs_his.nc` file; `structure_x`, `structure_y` & `structure_height`
- Option to include viscosity, enabling running on `theta=1.0`, with `viscosity = 1`. The values 'nuisc' will be automatically determined based on your grid resolution, and written to the log screen. Value can still be overruled by specifying 'nuisc = value' directly, or increased with e.g. a factor 2 using 'nuiscdim = 2'.
- Save maximum velocity proxy (in m/s) on 'dtmaxout' interval: `storevelmax =`

interval: storeqdrain = 1

- Bugfix in weir formulation
- Updated documentation
- Added tests in skillbed report
- Compliance with new Python setup tools HydroMT-SFINCS release v1.0.1

Official open source version: v2.0.0 Alpe d'Huez release

On the 16th of November 2022, we have made SFINCS open source available as the SFINCS v2.0.0 Alpe d'Huez release, 'Moving Dutch Mountains in compound flood modelling'. This contains open access to the source code and executables from Github: <https://github.com/Deltares/SFINCS>. The code consists of all functionality of v1, with the large addition of the subgrid mode and first GPU functionality using openacc. For more details, see below.

Pre-release version(s): v1 revision XXX

Before making SFINCS open source, version history was controlled using subversion numbering. Therefore papers using pre-release versions of SFINCS for instance refer to 'trunk revision 141', as in Leijnse et al. 2021. These version 1 revisions contained all standard SFINCS functionality for the regular mode.

Recent advancements in accuracy: subgrid mode

Hereby some examples regarding subgrid features and GPU computing.

What are subgrid features?

Subgrid features are a method in which flux computations are performed on a coarser grid than the update of the water levels which is done on a much finer resolution. In this way computations can be sped up, while still using high resolution information of topography and bathymetry.

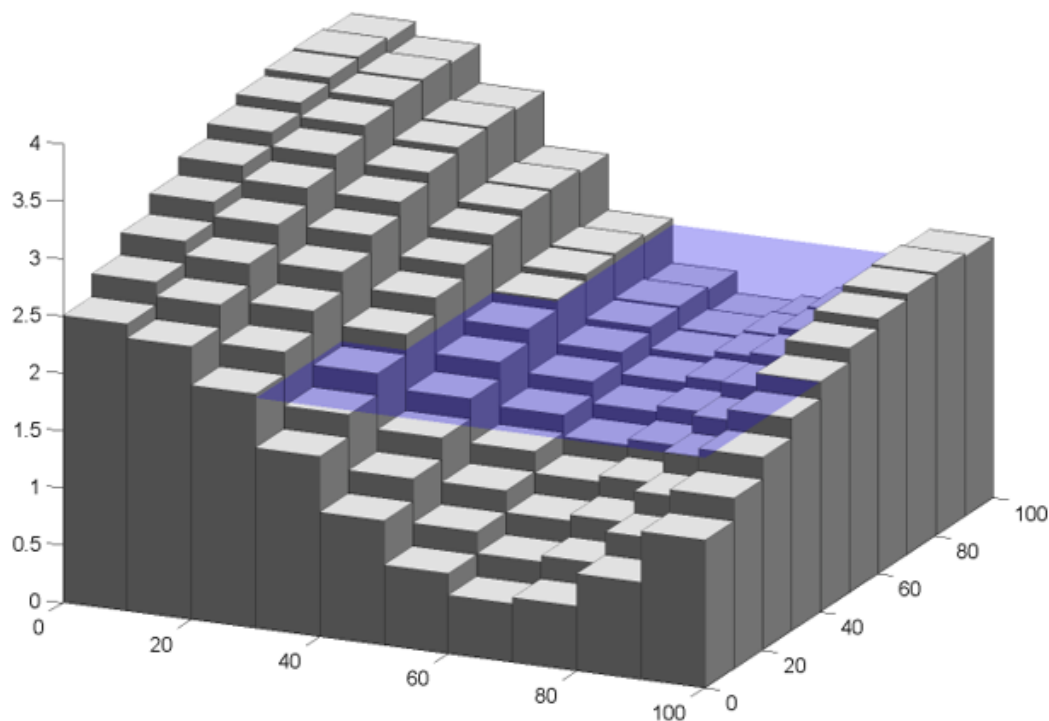


Fig. 11 Example subgrid features within one grid cell

Why subgrid features?

Often model runtimes are too large to go to very fine resolution modelling because refining a grid size with a factor 2, leads to a 2^3 longer model runtime due to the time step limitation in the CFL-criteria. This can be overcome by using a subgrid approach for the continuity update. This has the benefit that larger grid domains can be used while keeping accurate results.

How does it work?

The subgrid method implemented so that subgrid tables are derived in pre-processing that contain relations between the water level and volume for every grid cell. These tables are derived using high resolution topography and bathymetry data. In the SFINCS model itself, these subgrid tables are used to determine an accurate estimation of the water level after calculating fluxes on a coarser grid resolution. Additionally, for calculating the fluxes between cells, a representative water depth is determined. This makes it possible to compute on a coarser grid resolution (improvement of efficiency) while still detailed information about the local elevation is incorporated when determining corresponding water levels leading to accurate results.

Increase in computational efficiency?

Is your app OWASP compliant? Get the latest guidelines for implementing robust mobile app security.

 **stable** ▼

Ads by
EthicalAds

Close Ad

grid, the computational speedup is a factor 8. Our case study in Houston shows that even larger increases in speed are possible! See: <https://agu2020fallmeeting-agu.ipostersessions.com/Default.aspx?s=9C-05-18-CF-F1-2B-17-F0-7A-21-93-E6-13-AE-F3-24>

Recent advancements in speed: GPU enabled

The SFINCS source code has now been GPU enabled to make optimal use of fast Graphics Processing Unit computers. For more information get in touch with us!